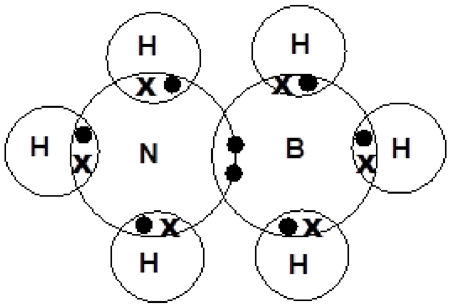
SECTION B

Question				Marking details	Marks available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
6	(a)	(i)		base is a proton acceptor (1) lone pair on nitrogen (can form coordinate bond with H^+) (1)	2			2		
		(ii)	I	award (1) for any of following ammonium salt of any strong acid e.g. ammonium sulfate, ammonium nitrate, ammonium chloride any strong acid e.g. sulfuric acid, nitric acid, hydrochloric acid	1			1		
			II	award (1) for any sensible answer e.g. using enzymes storage of biological molecules fermentation	1			1		1
	(b)	(i)		 accept correct versions using triangles (or other symbols) to differentiate between electrons	1			1		

<table>
<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td></td><td></td><td>(ii)</td><td></td><td>award (2) for all three boxes correct
award (1) for any two boxes correct
<table>
<tr><th></th><th>Similarity between hexagonal boron nitride and graphite</th><th>Difference between hexagonal boron nitride and graphite</th></tr>
<tr><td>Structure</td><td>layers of hexagonal sheets present in both</td><td>layers in register in boron nitride and out of register in graphite / atoms of adjacent layers above one another in boron nitride but not in graphite</td></tr>
<tr><td>Bonding</td><td>each atom is bonded by covalent bonds to three others in both graphite and BN</td><td>delocalised electrons in graphite but not in boron nitride / bonds are non-polar in graphite but polar in boron nitride</td></tr>
</table></td><td>2</td><td></td><td></td><td>2</td><td></td><td></td></tr>
<tr><td></td><td>(c)</td><td></td><td></td><td>phosphorus has available / accessible d-orbitals (1)
whilst nitrogen does not (1)

or

phosphorus can expand its octet (1)
but nitrogen cannot (1)</td><td>2</td><td></td><td></td><td>2</td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td></td><td>Question 6 total</td><td>9</td><td>0</td><td>0</td><td>9</td><td>0</td><td>0</td></tr>
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